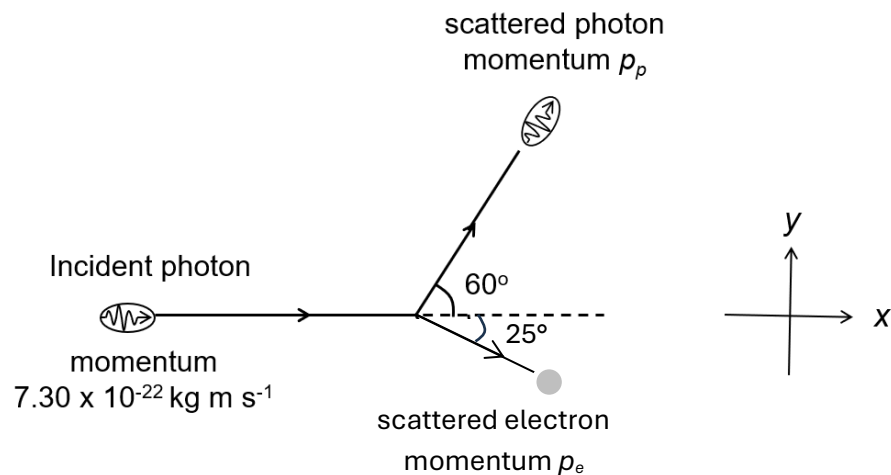


- 1 Fig. 1.1 shows an incident photon of momentum  $7.30 \times 10^{-22} \text{ kg m s}^{-1}$  colliding with a stationary electron.



**Fig. 1.1**

After the collision, the photon is scattered off through an angle of  $60^\circ$  and has a momentum  $p_p$ . The electron gets scattered off at an angle of  $25^\circ$  with a momentum  $p_e$ . Their scattering angles are measured with respect to the path of the incident photon.

- (a) Explain why linear momentum is conserved in this collision for the system of photon and electron.

.....  
 ..... [1]

- (b) Consider the photon and electron as a system.

- (i) State the total momentum of the system along the

1.  $x$ -direction,

momentum in  $x$ -direction = .....  $\text{kg m s}^{-1}$  [1]

2.  $y$ -direction.

momentum in  $y$ -direction = .....  $\text{kg m s}^{-1}$  [1]

- (ii) Applying the principle of conservation of momentum in both directions, determine the momentum  $p_e$  of the electron after the collision.

momentum  $p_e$  of the electron = ..... kg m s<sup>-1</sup> [3]

[Total: 6]